

Jacob Rosario

Software Engineer

Vancouver, BC | U.S. & Canadian Citizen | 604-230-8765 | jacobrosario0225@gmail.com | [LinkedIn](#) | [GitHub](#) | jacobrosario.com

Professional Summary

Early-career Software Engineer with hands-on experience building and deploying web and iOS applications using JavaScript, React, Swift, SwiftUI, Python, and Three.js. Developed real-time data and motion-sensing tools used by coaches and athletes across 20+ competitive matches. Skilled at turning complex data and user needs into responsive, practical software products.

Skills

- **Languages:** JavaScript, Python, Swift, HTML, CSS
- **Frameworks & Libraries:** React, SwiftUI, Three.js
- **Tools & AI:** Claude, ChatGPT, LLMs, Git, GitHub

Projects

RepPower | Swift, SwiftUI | [GitHub/RepPower](#) **Sep 2026**

- Developing an iOS application that uses AirPods motion-sensor data to estimate barbell squat velocity for velocity-based training.
- Engineered a Swift physics pipeline that transforms raw motion data into vertical acceleration and velocity while accounting for sensor noise and integration drift.
- Implemented zero-velocity drift correction and optimized high-frequency processing to maintain stable real-time measurements without blocking the SwiftUI interface.

Volleyball Stat Tracker | JavaScript, React | [GitHub/Volleyball-Stat-Tracker](#) **Feb 2026**

- Deployed a real-time React stat tracker used courtside on iPad across 20+ competitive matches, replacing handwritten tracking with live performance data.
- Refactored redundant stat sources into a single source of truth, eliminating state drift and supporting tactical decisions that contributed to the team's Division 3 → Division 2 climb

Spike Timer Analyzer | JavaScript, React | [Spike Timer \(Web App\)](#) **Mar 2026**

- Built a client-side video-analysis tool that measures spike timing from user-marked takeoff, contact, and landing frames.
- Selected a lightweight geometric approach instead of pose-detection machine learning, enabling responsive browser-based processing.

Interactive 3D Portfolio | React, Three.js | jacobrosario.com **Jun 2026**

- Engineered a responsive 3D portfolio featuring custom physics simulations, model rendering, lighting, and interactive environments.

Experience

Hustle Volleyball Club **Feb 2026 - May 2026**

Sports Analytics Developer & Coach

Burnaby & Vancouver, BC

- Integrated a custom JavaScript stat tracker into match preparation and live coaching workflows to inform lineups, practices, and in-game adjustments.
- Coached a 15-player competitive team through a 4-month season of 5+ tournaments, including provincials and nationals.
- Helped the team qualify for the national championship through data-informed preparation and coaching.

Vancouver College **Sep 2025 - Dec 2025**

Performance Analyst & Assistant Coach

Vancouver, BC

- Analyzed 5+ matches per week using video review and statistical data to identify opponent tendencies and performance patterns.
- Translated video data into actionable recommendations for lineups, opponent-specific strategies, and player development.

Starbucks **Sep 2023 - Sep 2025**

Barista

Vancouver, BC & San Francisco, CA

- Processed 100+ transactions per shift at high-volume locations while maintaining accuracy and service quality under pressure.

Education

University of British Columbia **Jun 2025**

Bachelor of Science (BSc), Biology

Vancouver, BC

- **Achievements:** Dean's List

Certifications

- Anthropic: Claude Code 101
- Anthropic: Claude 101
- Anthropic: AI Fluency: Frameworks & Foundations

Hobbies & Interests

Competitive volleyball, coaching & performance analysis | Sports analytics & biomechanics | AR/VR & 3D technology | Video games | Weightlifting & Olympic lifting | Content creation & video editing | Specialty coffee | Anime & manga | Playing guitar